

PRESCRIBING INFORMATION

NEO-PENOTRAN® FORTE (METRONIDAZOLE/ MICONAZOLE NITRATE) 750MG/200MG VAGINAL PESSARY

Active ingredients:

Each pessary contains 750 mg metronidazole and 200 mg miconazole nitrate.

Product Description:

NEO-PENOTRAN FORTE PESSARY is slightly white or yellowish ellipsoid pessary.

Pharmacodynamics:

NEO-PENOTRAN FORTE PESSARY contains miconazole nitrate for antifungal and metronidazole for antibacterial and antitrichomonal effects. Miconazole nitrate which is a synthetic imidazole derivative antifungal has a wide spectrum of activity and is particularly effective against pathogen fungi including *Candida albicans*. In addition, miconazole nitrate is effective against Gram positive bacteria. Miconazole shows its effect by ergosterol synthesis in the cytoplasmic membrane. Miconazole nitrate changes permeability of the mycotic cell of *Candida species* and inhibits glucose utilization *in vitro*.

Metronidazole, 5-nitroimidazole derivative is an antiprotozoal and an antibacterial agent and is effective against several infections caused by anaerobic bacteria and protozoa, such as *Trichomonas vaginalis*, *Gardnerella vaginalis* and anaerobic bacteria including anaerobic streptococci.

Miconazole nitrate and metronidazole have no synergic or antagonistic effect.

The clinical cure rates achieved in an open, monocenter, uncontrolled clinical study to assess the efficacy and safety of NEO-PENOTRAN FORTE PESSARY for 7 days in 104 patients with clinical / microbiological diagnosis of vaginitis are 96.6% candidal vulvovaginitis, 98.1% for bacterial vaginosis, 97.3% for trichomonal vaginitis and 98% for mixed vaginal infections. Microbiological cure rates are 89.8%, 96.2%, 100%, 91.7% for each infection respectively.

Pharmacokinetics:

General Properties:

Absorption:

Miconazole nitrate: Absorption of miconazole nitrate by the intravaginal route is very low (approximately 1.4% of dose). Following intravaginal application of NEO-PENOTRAN FORTE PESSARY, miconazole nitrate was not detected in plasma.

Metronidazole: Bioavailability of metronidazole by this route is 20% compared to the oral route. Steady state levels of metronidazole in plasma ranged from 1.1 to 5.0 µg/ml after daily intravaginal application of NEO-PENOTRAN FORTE PESSARY.

Distribution:

Miconazole nitrate: The protein binding is 90%-93%. Its distribution to cerebrospinal fluid is poor, however it is widely distributed to other tissues. Distribution volume is 1400 L.

Metronidazole: It is widely distributed in body tissues and fluids including bile, bone, breast, milk, cerebral abscesses, cerebrospinal fluid, liver and liver abscesses, saliva, seminal fluids and vaginal secretion, and achieves concentrations similar to those in plasma. It crosses the placenta and rapidly enters to fetal circulation. No more than 20% is bound to plasma proteins. Distribution volume is 0.25-0.85 L/kg.

Biotransformation:

Miconazole nitrate: It is metabolized in liver. Two non-active metabolites are found (2,4-dichlorophenyl-1 H imidazole ethanol and 2,4-dichloromandelic acid).

Metronidazole: It is metabolized in the liver by oxidation, hydroxy metabolite is active. Major metabolites of metronidazole, hydroxy and acetic acid metabolites, are excreted in urine. The hydroxy metabolite has a 30% of biological activity of metronidazole.

Elimination:

Miconazole nitrate: Half-life is 24 hours. Less than 1% is excreted in the urine. Approximately 50%, usually unchanged, excreted by faeces.

Metronidazole: Half-life is 6-11 hours. About 6%-15% of metronidazole dose is excreted with faeces, 60%-80% is unchanged and excreted in urine as its metabolites. Approximately 20% of metronidazole is excreted in the urine as unchanged drug.

Indication:

NEO-PENOTRAN FORTE PESSARY is used in the treatment of candidal vulvovaginitis due to *Candida albicans*, in bacterial vaginosis due to anaerobic bacteria and *Gardnerella vaginalis*, in trichomonal vaginitis due to *Trichomonas vaginalis* and in mixed vaginal infections.

Recommended Dose:

Do not use without consulting a physician. If it is not advised otherwise by a physician; one pessary should be inserted high into the vagina at night for 7 days. In recurrent cases, application of one pessary at night for 14 days is recommended. It is not recommended that the administration of NEO-PENOTRAN FORTE PESSARY in the menstrual period because of the impaired efficacy of the product or facing some difficulties during administration.

Additional information on special populations:

Renal/Liver failure: In renal failure, the half-life of metronidazole is not changed. Therefore, dose reduction is not necessary but in seriously impaired renal function which hemodialysis is required, dose adjustment should be done.

In advanced hepatic dysfunctions metronidazole clearance may be impaired. Metronidazole may increase encephalopathy symptoms due to increased plasma levels and therefore should be used carefully in hepatic encephalopathy patients.

Paediatric population: Not to be used in children under 18.

Geriatric population: Same dose as for adults is administered in elderly over 65 years.

Method of Administration:

Only for intravaginal use.

NEO-PENOTRAN FORTE PESSARY should be applied deep into the vagina, in a supine position. Not to be swallowed or applied by other routes.

Contraindications:

NEO-PENOTRAN FORTE PESSARY should not be used:

- In patients known to be hypersensitive to the active ingredients or their derivatives
- In patients using alcohol during the treatment or within 3 days after the treatment
- In patients using disulfiram during the treatment or within 2 weeks
- During the first trimester of pregnancy
- In pregnant women who have trichomonal vaginitis in the first trimester
- In cases of porphyria, epilepsy, advanced hepatic dysfunction.

Warnings and Precautions:

Patients should be advised not to take alcohol during the therapy and for 3 days after the end of the treatment, because of the possibility of a disulfiram-like reaction.

High doses and long-term metronidazole may cause peripheral neuropathy and convulsion due to systemic usage.

Should not be used in virgins and in young people (girls) who are not sexually mature.

The pessary must not be used with contraceptive diaphragms and preservatives since the pessary base may react with the rubber.

Other vaginal products (e.g. tampon, douche or spermicide) should not be used concurrently during the treatment.

Sexual partners with *Trichomonas vaginalis* should be treated at the same time.

Cases of severe hepatotoxicity/ acute hepatic failure, including cases with a fatal outcome with very rapid onset after treatment initiation in patients with Cockayne syndrome have been reported with products containing metronidazole for systemic use. In this population, metronidazole should therefore be used after careful benefit-risk assessment and only if no alternative treatment is available. Liver function tests must be performed just prior to the start of therapy, throughout and after end of treatment until liver function is within normal ranges, or until the baseline values are reached. If the liver function tests become markedly elevated during treatment, the drug should be discontinued.

Patients with Cockayne syndrome should be advised to immediately report any symptoms of potential liver injury to their physician and stop taking metronidazole.

Please consult your physician/ pharmacist before using this product if you are on the anticoagulant medicine warfarin, because bleeding from nose/ gums or bruising may occur spontaneously.

Sila dapatkan nasihat doktor atau ahli farmasi sebelum menggunakan keluaran ini jika anda mengambil ubat warfarin, iaitu sejenis ubat antipembekuan darah, kerana lebam/ pendarahan pada gusi/ hidung boleh berlaku secara spontan.

Interactions with Other Medicaments:

Due to metronidazole absorption, the following interactions can be seen if used concomitantly with the drugs below:

Alcohol: Alcohol intolerance (disulfiram - like reaction)

Amiodaron: Increase in risk of cardiotoxicity (QT prolongation, torsades de pointes, cardiac arrest)

Astemizole and terfenadine: Metronidazole inhibits the metabolism of these drugs and increases their plasma concentrations

Carbamazepine: Increase in blood concentration of carbamazepine

Cimetidine: Increase in blood levels of metronidazole and the risk of neurologic side effects

Cyclosporine: Increase in cyclosporine toxicity risk

Disulfiram: Central nervous system related effects (e.g. psychotic reactions)

Lithium: Increase in blood levels and lithium toxicity
Phenytoin: Increase in blood levels of phenytoin, decrease in blood levels of metronidazole
Phenobarbital: Decrease in blood levels of metronidazole
Fluorouracil: Increase in blood levels and toxicity of fluorouracil
Oral anticoagulants: Increase in anticoagulant effect.

Interference with blood levels of liver enzymes, glucose (hexokinase method), theophylline and procainamide have been observed during the treatment with metronidazole.

Due to miconazole nitrate absorption, the following interactions can be seen if used concomitantly with the drugs below:

Acenocoumarol, Anisidione, Dicumarol, Phenindione, Phenprocoumon, Warfarin: Increase in bleeding risk
Astemizole, cisaprid and terfenadine: Miconazole inhibits the metabolism of these drugs and increases their plasma concentrations
Carbamazepine: Decrease in carbamazepine metabolism
Cyclosporine: Increase in cyclosporine risk toxicity (renal dysfunction, cholestasis, parasthesias)
Fentanyl: Increase or prolonged effects of opioid (CNS depression, respiratory depression)
Phenytoin and Fosphenytoin: Increase in phenytoin toxicity risk (ataxia, hyperreflexia, nystagmus, tremor)
Glimepiride: Hypoglycemia
Oxybutinin: Increase in plasma concentration or exposure to oxybutinin as a result of inhibiting the metabolism of oxybutynin (dry mouth, constipation, headache)
Oxycodone: Increase in oxycodone plasma concentration and reduction in clearance
Pimozide: Increase in cardiotoxicity risk (QT prolongation, torsades de pointes, cardiac arrest)
Tolterodine: Increase in tolterodine bioavailability in individuals with deficient cytochrome P450 2D6 activity
Trimetrexate: Increase in trimetrexate toxicity (bone marrow suppression, renal and hepatic dysfunction and gastrointestinal ulceration).

Additional information on special populations: No interaction study has been conducted on special populations.

Paediatric population: No interaction study has been conducted on children.

Pregnancy and Lactation:

General Advice: Pregnancy category is C.

Women of child bearing potential / Birth Control (Contraception): Since the effects of active ingredients in NEO-PENOTRAN FORTE PESSARY for fetus and newborn growth are not clearly known, women who must use this product should avoid pregnancy with a proper birth control method.

Pregnancy: Preclinical studies on animals regarding the pregnancy, embryonal/fetal growth, perinatal and/or postnatal growth are insufficient. Potential risk for human is not known. There is insufficient data regarding the use of NEO-PENOTRAN FORTE PESSARY in the first trimester of pregnancy. Therefore, NEO-PENOTRAN FORTE PESSARY should not be used in the first trimester of pregnancy. In the second and third trimesters, benefit/risk ratio should be evaluated by a physician, should not be used unless it is necessary.

Lactation: Breastfeeding should be discontinued, since metronidazole appears in milk. Breastfeeding can be started again 24-48 hours after the end of treatment.

Reproduction / Fertility: There is no evidence regarding hazardous effect on human and animal fertility when metronidazole or miconazole nitrate is administered alone.

Side Effects:

The frequency of adverse events listed below is defined using the following convention:
Very common (≥1/10); common (≥1/100 to <1/10); uncommon (≥1/1,000 to <1/100), rare (≥1/10,000 to <1/1,000); very rare (<1/10,000), not known (cannot be estimated from the available data).

The frequency of systemic side/adverse effects is very rare since after intravaginal administration of metronidazole, very low plasma levels are observed (2% - 12% compared to oral route).

Miconazole nitrate can cause vaginal irritation (burning, itching) as all other imidazole derivative antifungal drugs applied intravaginally (2-6%). In vaginitis, vaginal mucosa could be inflamed, therefore vaginal burning and itching may see after the first pessary is applied or toward to the third day of the treatment. These symptoms are fastly reduced while the treatment continues. If there is a severe irritation symptoms, the treatment should be stopped.

Undesirable effects regarding systemic use of active substances of NEO-PENOTRAN FORTE PESSARY are listed below:

Blood and lymphatic system disorders:

Not known: Leukopenia

Immune system disorders:

Not known: Hypersensitivity reactions, allergic reactions (anaphylaxis may occur in severe cases)

Psychiatric disorders:

Uncommon: Depression

Very rare: Mental alterations.

Nervous system disorders:

Common: Dizziness, headache

Not known: Tiredness or weakness, ataxia, convulsion, peripheral neuropathy due to intensive and/or prolonged metronidazole therapy.

Gastrointestinal disorders:

Not known: Taste changes, metallic taste, nausea, vomiting, constipation, dry mouth, diarrhea, lack of appetite, abdominal pain or cramp.

General disorders and administration site conditions:

Very common: Vaginal discharge

Common: Vaginitis, vulvovaginal irritation, pelvic discomfort

Uncommon: Feeling of thirst

Rare: Vaginal burning, itching, irritation, stomachache, rash

Not known: Local irritation and hypersensitivity, contact dermatitis.

These adverse effects are observed rarely since blood concentration of metronidazole is much lower when administered via intravaginal route.

Symptoms and Treatment of Overdose:

When excessive amount of pessary is applied, systemic effects may occur due to metronidazole, but intravaginal metronidazole is not expected to cause life-threatening symptoms.

Symptomatic and supportive treatment should be instituted. There is no specific antidote for metronidazole. Cure can be provided in individuals who ingested a dose of 12 g of metronidazole.

Symptoms of metronidazole overdose are nausea, vomiting, abdominal pain, diarrhea, itching, metallic taste, ataxia, dizziness, paresthesia, convulsion, leukopenia, darkening of urine; symptoms of miconazole nitrate overdose are sore mouth and throat, anorexia, nausea, vomiting, headache, diarrhea.

Effects on Ability to Drive and Use Machine:

Systemic administration of metronidazole may affect driving and using machines. In comparison with systemic use, topical metronidazole is absorbed in vagina in low concentrations.

NEO-PENOTRAN FORTE PESSARY may cause dizziness, ataxia, fatigue and weakness, therefore may affect driving or operating machinery.

PreClinical Data Safety:

Preclinical data pharmacology, repeated dose toxicity, genotoxicity, carcinogenic potential, reproduction toxicity studies reveal no special hazard for humans.

In microbiological in vitro study, there is no synergic or antagonistic interactions against *Candida albicans*, *Streptococcus* (gram B of Lancefield), *Cardnerella vaginalis* and *Trichomonas vaginalis* between active substances of the combination.

A preclinical study carried out with the combination of 750 mg metronidazole and 200 mg miconazole nitrate indicated no potentiation or synergism with regards to lethal or toxic effects of the ingredients in female rats.

In a vaginal mucous irritation study in female Beagle dogs with the same combination, it was concluded that it does not cause vaginal mucous irritation as well as nor clinical, biochemical and haematological alterations. In the same study, local and systemic toxic effects are not detected.

Shelf life:

36 months

Storage:

Store at room temperature below 30°C.

Packaging:

PVC/LDPE duplex foil casing strips

Contents of container: 7 vaginal pessary.

Manufacturer:

Exeltis Ilac San. Ve Tic. A.S.

Cerkezoy Organize Sanayi Bolgesi,

Gaziosmanpasa Mahallesi,

Fatih Bulvari,

No:19/2 Cerkezoy-Tekirdag,

Turkey.

Product Registration Holder:

Eurodrug Laboratories (M) Sdn Bhd

T2-21-10, THE SQUARE, ONE CITY,

Jalan USJ 25/1C, 47650 Subang Jaya,

Selangor, Malaysia.

Date of Revision: 28 Nov 2024